Expected value/Mathematical expectation and variance of Random variable :

Example: 4.1

Mean= u = E(X) = ∑ x f(x) = 1.7

$Var(X) = E(X^2) – [E(X)]^2$

Where $E(X^2) = \sum X^2 f(x) = 0 + 1^2 (12/35) + 2^2 (18/35) + 3^2 (4/35) = 3.4285$

So, $Var(X)= 3.4285 – (1.7)^2 = 0.5385$

Sd(X)= $\sqrt{0.5385}$ = 0.7338

Expected Value/Mathematical expectation and Variance of Function of random variable:

Example: 4.4

g(x) = 2x-1 (where g(x) is a function of X) (by property E(2x-1) = 2E(X)-1)

E(g(x)) = ∑ g(x) f(x) = ∑ (2x-1) f(x) = 12.67 $

$Var(g(x)) = E(g(x)^2) – [E(g(x)]^2$

$E(g(x)^2) = E (2X-1)^2 f(x)$

$= \sum (2X-1)^2 f(x)$

= 49 (1/12) + 81 (1/12) + 121 (1/4)………. 289 (1/6)

= 169.0

$Var(g(x)) = 169.0 – (12.67)^2$ (by property Var (2x-1) = 4 var (x))

= 8.4711

Sd(g(x)) = $\sqrt{8.4711} = 2.91$

Properties of Mathematical Expectation:

- E(a) = a , where ‘a’ is a constant
- E(aX+b) = a E(X) + E(b) = a E(X) + b where ‘a’ and ‘b’ are constants
- E(X+Y) = E(X) +E(Y) where ‘X’ and ‘Y’ are two random variables
- E(X-Y) = E(X) – E(Y)
- E(XY) = E(X) E(Y)

Properties of Variance:

- Var(a) = 0, where ‘a’ is a constant
- $Var(ax+b) = a^2 var(x) + 0 = a^2 var(x)$ where ‘a’ and ‘b’ are constants
- Var(x+y) = var(x) + var(y) + 2cov (x,y) (If variables are independent then cov(x,y) = 0)
- Var (x-y) = var(x) + var(y) – 2cov (x,y)